

YARA Setup Into Wazuh

```
sudo apt update
sudo apt install -y make gcc autoconf libtool libssl-dev pkg-config jq
sudo curl -LO https://github.com/VirusTotal/yara/archive/v4.5.5.tar.gz
sudo tar -xvzf v4.5.5.tar.gz -C /usr/local/bin/ && rm -f v4.5.5.tar.gz
cd /usr/local/bin/yara-4.5.5/
sudo ./bootstrap.sh && sudo ./configure && sudo make && sudo make install
sudo ldconfig
```

Untuk instalasi YARA dibutuhkan semacam kumpulan script & package2 untuk membantu dalam detect malware behavior.

```
libtool: install: /usr/bin/install -c yara /usr/local/bin/yara
libtool: install: /usr/bin/install -c .libs/yarac /usr/local/bin/yarac
/usr/bin/mkdir -p '/usr/local/include'
/usr/bin/install -c -m 644 libyara/include/yara.h '/usr/local/include'
/usr/bin/mkdir -p '/usr/local/share/man/man1'
/usr/bin/install -c -m 644 'yara.man' '/usr/local/share/man/man1/yara.1'
/usr/bin/install -c -m 644 'yarac.man' '/usr/local/share/man/man1/yarac.1'
/usr/bin/mkdir -p '/usr/local/lib/pkgconfig'
/usr/bin/install -c -m 644 yara.pc '/usr/local/lib/pkgconfig'
/usr/bin/mkdir -p '/usr/local/include/yara'
/usr/bin/install -c -m 644 libyara/include/yara/ahocorasick.h libyara/include/yara/arena.h libyara/include/yara/atoms.h libyara/include/yara/base64.h libyara/include/yara/bitmask.h libyara/include/yara/compiler.h libyara/include/yara/error.h libyara/include/yara/exe.h libyara/include/yara/exefiles.h libyara/include/yara/filemap.h libyara/include/yara/hash.h libyara/include/yara/integers.h libyara/include/yara/libyara.h libyara/include/yara/limits.h libyara/include/yara/mem.h libyara/include/yara/modules.h libyara/include/yara/notebook.h libyara/include/yara/object.h libyara/include/yara/parser.h libyara/include/yara/proc.h libyara/include/yara/re.h libyara/include/yara/rules.h libyara/include/yara/scan.h libyara/include/yara/scanner.h libyara/include/yara/simple_str.h libyara/include/yara/sizedstr.h libyara/include/yara/stack.h libyara/include/yara/stopwatch.h libyara/include/yara/stream.h libyara/include/yara/strut.h libyara/include/yara/threading.h libyara/include/yara/types.h libyara/include/yara/unaligned.h libyara/include/yara/utls.h '/usr/local/include/yara'
make[2]: Leaving directory '/usr/local/bin/yara-4.5.5'
make[1]: Leaving directory '/usr/local/bin/yara-4.5.5'
/usr/local/lib
a6@6-agent: /usr/local/bin/yara-4.5.5$ echo "/usr/local/lib" | sudo tee -a /etc/ld.so.conf
sudo ldconfig
yara --version      # harus keluar 4.5.5
/usr/local/lib
4.5.5
a6@6-agent: /usr/local/bin/yara-4.5.5$ sudo mkdir -p /tmp/yara/rules
sudo curl 'https://valhalla.nexttron-systems.com/api/v1/get' \
```

Agent (IP Agent1)

Setup YARA.sh

```
1 sudo nano /var/ossec/active-response/bin/yara.sh
```

```
#!/bin/bash
# Wazuh active response: scan file yang berubah dengan YARA
LOG_FILE="/var/ossec/logs/active-responses.log"

read INPUT_JSON
YARA_PATH=$(echo "$INPUT_JSON" | jq -r .parameters.extra_args[1])
YARA_RULES=$(echo "$INPUT_JSON" | jq -r .parameters.extra_args[3])
FILENAME=$(echo "$INPUT_JSON" | jq -r .parameters.alert.syscheck.path)

if [[ -z "$YARA_PATH" || -z "$YARA_RULES" ]]; then
    echo "wazuh-yara: ERROR - yara_path/yara_rules kosong." >> "$LOG_FILE"
fi
if [[ ! -e "$FILENAME" ]]; then
    echo "wazuh-yara: ERROR - file hilang: $FILENAME" >> "$LOG_FILE"; exit 1
fi

# tunggu file selesai ditulis (size stabil) biar tidak scan file parsial
size=0; actual=$(stat -c %s "$FILENAME" 2>/dev/null)
while [[ "$size" != "$actual" ]]; do sleep 1; size=$actual; actual=$(stat -c %s "$FILENAME" 2>/dev/null)

# scan: flag -d (external var kosong) + 2>/dev/null WAJIB, karena signal
yara_output="$("$YARA_PATH/yara" -w -d filename="" -d filepath="" -d rules "$YARA_RULES" 2>/dev/null)"

if [[ -n "$yara_output" ]]; then
    while read -r line; do
        echo "wazuh-yara: INFO - Scan result: $line" >> "$LOG_FILE"
    done <<< "$yara_output"
fi
exit 0
```

Cek version dulu

```
echo "/usr/local/lib" | sudo tee -a /etc/ld.so.conf && sudo ldconfig
yara --version      # harus keluar: 4.5.5
```

Nextron menyediakan database Malware Threat Detection signature/behavior, jadi ini dapat menjadi opsi tambahan selain custom Rules

Terdapat demo api yang memang disediakan oleh Nextron sehingga ini dapat membantu secara public.

[illegible]

(Demo API Key dapat ditemukan di website official)

```
sudo apt install -y git
sudo mkdir -p /tmp/yara/rules
sudo git clone --depth 1 https://github.com/Neo23x0/signature-base.git
cat /tmp/yara/signature-base/yara/*.yar | sudo tee /tmp/yara/rules/yara_rules.yar
grep -c "^rule " /tmp/yara/rules/yara_rules.yar

sudo cp /tmp/yara/rules/yara_rules.yar /var/ossec/yara/rules/yara_rules.yar

sudo chown -R root:wazuh /var/ossec/yara
sudo chmod -R 750 /var/ossec/yara
```

```
sudo mkdir -p /tmp/yara/malware

curl -sL "https://raw.githubusercontent.com/wazuh/wazuh-documentation/1.15.0/
/usr/local/bin/yara -w -r /tmp/yara/rules/yara_rules.yar /tmp/yara/malw
```

```
sudo mkdir -p /opt/malware-samples
sudo chown root:wazuh /opt/malware-samples
sudo chmod 770 /opt/malware-samples
```

(Untuk load db/behavior)

```
/usr/local/bin/yara -w -d filename="" -d filepath="" -d extension="" -c
-r /tmp/yara/rules/yara_rules.yar /tmp/yara/malware/webshell
```

```
sudo cp /tmp/yara/malware/webshell /opt/malware-samples/webshell
/usr/local/bin/yara -w -d filename="" -d filepath="" -d extension="" -c
-r /var/ossec/yara/rules/yara_rules.yar /opt/malware-samples/webshell
```

```
sudo chown root:wazuh /var/ossec/active-response/bin/yara.sh
sudo chmod 750 /var/ossec/active-response/bin/yara.sh
```

- 1 sudo systemctl restart wazuh-agent
- 2 sudo systemctl status wazuh-agent --no-pager

Manager

```
sudo nano /var/ossec/etc/ossec.conf
```

```
<directories realtime="yes">/opt/malware-samples</directories>
```

```
sudo nano /var/ossec/etc/ossec.conf
```

```
<command>
  <name>yara_linux</name>
  <executable>yara.sh</executable>
  <extra_args>-yara_path /usr/local/bin -yara_rules /var/ossec/yara/rules</extra_args>
  <timeout_allowed>no</timeout_allowed>
</command>
<active-response>
  <command>yara_linux</command>
  <location>local</location>
  <rules_id>100300,100301</rules_id>
</active-response>
```

```
sudo nano /var/ossec/etc/decoders/local_decoder.xml
```

```
<decoder name="yara_decoder"><prematch>wazuh-yara:</prematch></decoder>
<decoder name="yara_decoder1">
  <parent>yara_decoder</parent>
  <regex>wazuh-yara: (\S+) - Scan result: (\S+) (\S+)</regex>
  <order>log_type, yara_rule, yara_scanned_file</order>
</decoder>
```

```
sudo nano /var/ossec/etc/rules/local_rules.xml
```

```
<group name="syscheck,">
  <rule id="100300" level="7"><if_sid>550</if_sid><field name="file">/c
    <description>File modified in monitored dir.</description></rule>
```

```

    <rule id="100301" level="7"><if_sid>554</if_sid><field name="file">/c
      <description>File added to monitored dir.</description></rule>
</group>
<group name="yara,">
  <rule id="108000" level="0"><decoded_as>yara_decoder</decoded_as><des
  <rule id="108001" level="12"><if_sid>108000</if_sid>
    <match>wazuh-yara: INFO - Scan result: </match>
    <description>YARA match: "$(yara_scanned_file)" rule $(yara_rule)</
</group>

```

```

sudo systemctl restart wazuh-manager
sudo systemctl status wazuh-manager --no-pager

```

Testing

(Di manager)

```

a6@a6-manager:~$ echo "wazuh-yara: INFO - Scan result: EICAR_Test /opt/
Starting wazuh-logtest v4.7.5
Type one log per line

**Phase 1: Completed pre-decoding.
    full event: 'wazuh-yara: INFO - Scan result: EICAR_Test /opt/me

**Phase 2: Completed decoding.
    name: 'yara_decoder'
    log_type: 'INFO'
    yara_rule: 'EICAR_Test'
    yara_scanned_file: '/opt/malware-samples/x'

**Phase 3: Completed filtering (rules).
    id: '108001'
    level: '12'
    description: 'YARA match: "/opt/malware-samples/x" rule EICAR_1
    groups: '['yara']'

```

```
firedtimes: '1'
mail: 'True'
**Alert to be generated.
```

(Di Agent)

```
sudo cp /opt/malware-samples/webshell /opt/malware-samples/uji_$(date +%Y%m%d_%H%M%S)
sudo tail -f /var/ossec/logs/active-responses.log

2026/06/04 12:04:25 active-response/bin/firewall-drop: {"version":1,"os":1}
2026/06/04 12:04:25 active-response/bin/firewall-drop: {"version":1,"os":1}

2026/06/04 12:04:25 active-response/bin/firewall-drop: Aborted
2026/06/04 12:05:26 active-response/bin/firewall-drop: Starting
2026/06/04 12:05:26 active-response/bin/firewall-drop: {"version":1,"os":1}

2026/06/04 12:05:26 active-response/bin/firewall-drop: Ended
Wed Jun 24 16:37:48 UTC 2026 active-response/bin/restart.sh agent
wazuh-yara: INFO - Scan result: EXT_WEBSHELL_PHP_Generic /opt/malware-samples/EXT_WEBSHELL_PHP_Generic
wazuh-yara: INFO - Scan result: webshell_Worse_Linux_Shell_1 /opt/malware-samples/webshell_Worse_Linux_Shell_1
wazuh-yara: INFO - Scan result: Worse_Linux_Shell_php /opt/malware-samples/Worse_Linux_Shell_php
wazuh-yara: INFO - Scan result: WebShell_Worse_Linux_Shell_2 /opt/malware-samples/WebShell_Worse_Linux_Shell_2
```

Reference

- <https://github.com/virustotal/yara>
- <https://valhalla.nextron-systems.com/>

